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**DR. A.P.J. ABDUL KALAM TECHNICAL  
UNIVERSITY UTTAR PRADESH, LUCKNOW**



**EVALUATION SCHEME & SYLLABUS  
FOR**

**B.Tech. (8<sup>th</sup> Semester)  
OPEN ELECTIVES- III & IV**

**AS PER**

**AICTE MODEL CURRICULUM & NEP2020  
[Effective from the Session: 2025-26]**

**Note:**

1. The Student shall choose an open Elective from the list in such a manner that he/she has not studied the same course in any form during the degree programme.
2. \*\* It is mandatory that for these subjects (BOE090, BOE100 & BOE101) only Trained Faculty (who had done the FDP for these courses) will teach the courses.

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|      | <b>BOE086: QUALITY MANAGEMENT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |
|      | <b>DETAILED SYLLABUS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>3-1-0</b>     |
| Unit | Topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Proposed Lecture |
| I    | Quality Concepts: Evolution of Quality Control, concept change, TQM Modern concept, Quality concept in design, Review of design, Evolution of proto type. Control on Purchased Product: Procurement of various products, evaluation of supplies, capacity verification, Development of sources, procurement procedure. Manufacturing Quality: Methods and techniques for manufacture, inspection and control of product, quality in sales and services, guarantee, analysis of claims. | 08               |
| II   | Quality Management: Organization structure and design, quality function, decentralization, designing and fitting, organization for different type products and company, economics of quality value and contribution, quality cost, optimizing quality cost, seduction program. Human Factor in quality Attitude of top management, cooperation of groups, operators attitude, responsibility, causes of apparatus error and corrective methods.                                        | 08               |
| III  | Control Charts, Theory of control charts, measurement range, construction and analysis of R charts, process capability study, use of control charts. Attributes of Control Chart, Defects, construction and analysis of charts, improvement by control chart, variable sample size, construction and analysis of C charts                                                                                                                                                              | 08               |
| IV   | Defects diagnosis and prevention defect study, identification and analysis of defects, correcting measure, factors affecting reliability, MTTF, calculation of reliability, building reliability in the product, evaluation of reliability, interpretation of test results, reliability control, maintainability, zero defects, quality circle.                                                                                                                                        | 08               |
| V    | ISO-9000 and its concept of Quality Management, ISO 9000 series, Taguchi method, JIT in some details.                                                                                                                                                                                                                                                                                                                                                                                  | 08               |

Text Books:

1. Lt. Gen. H. Lal, "Total Quality Management", Eastern Limited.
2. Greg Bounds, "Beyond Total Quality Management", McGraw Hill
3. Menon, H.G, "TQM in New Product manufacturing", McGraw Hill

| <b>BOE093 AUTOMATION AND ROBOTICS</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  |
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| <b>DETAILED SYLLABUS</b>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>3-1-0</b>     |
| Unit                                  | Topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Proposed Lecture |
| I                                     | Automation: Definition, Advantages, goals, types, need, laws and principles of Automation. Elements of Automation. Fluid power and its elements, application of fluid power, Pneumatics vs. Hydraulics, benefit and limitations of pneumatics and hydraulics systems, Role of Robotics in Industrial Automation.                                                                                                                                                                                                                                                                  | 08               |
| II                                    | Manufacturing Automation: Classification and type of automatic transfer machines; Automation in part handling and feeding, Analysis of automated flow lines, design of single model, multimode and mixed model production lines. Programmable Manufacturing Automation CNC machine tools, Machining centers, Programmable robots, Robot time estimation in manufacturing operations.                                                                                                                                                                                              | 08               |
| III                                   | Robotics: Definition, Classification of Robots - Geometric classification and Control classification, Laws of Robotics, Robot Components, Coordinate Systems, Power Source. Robot anatomy, configuration of robots, joint notation schemes, work volume, manipulator kinematics, position representation, forward and reverse transformations, homogeneous transformations in robot kinematics, D-H notations, kinematics equations, introduction to robot arm dynamics.                                                                                                          | 08               |
| IV                                    | Robot Drives and Power Transmission Systems: Robot drive mechanisms: Hydraulic/Electric/Pneumatics, servo & stepper motor drives, Mechanical transmission method: Gear transmission, Belt drives, Rollers, chains, Links, Linear to Rotary motion conversion, Rotary-to-Linear motion conversion, Rack and Pinion drives, Lead screws, Ball Bearings. Robot end Effectors: Classification of End effectors – active and passive grippers, Tools as end effectors, Drive system for grippers. Mechanical, vacuum and magnetic grippers. Gripper force analysis and gripper design. | 08               |
| V                                     | Robot Simulation: Methods of robot programming, Simulation concept, Off-line programming, advantages of offline programming. Robot Applications: Robot applications in manufacturing-Material transfer and machine loading/unloading, Processing operations like Welding & painting, Assembly operations, Inspection automation, Limitation of usage of robots in processing                                                                                                                                                                                                      | 08               |

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|  | operation. Robot cell design and control, Robot cell layouts-<br>Multiple robots & Machine interference. |  |
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### **Text Books:**

1. An Introduction to Robot Technology, by Coifet Chirroza, Kogan Page.
2. Robotics for Engineers, by Y. Koren, McGraw Hill.
3. Robotic: Control, Sensing, Vision and Intelligence, by Fu, McGraw Hill.
4. Introduction to Industrial Robotics, by Nagrajan, Pearson India.
5. Robotics, by J.J. Craig, Addison-Wesley.
6. Industrial Robots, by Groover, McGraw Hill.
7. Robotic Engineering - An Integrated Approach : Richard D. Klafter  
Thomas A.
8. Robots & Manufacturing Automation, by Asfahl, Wiley.